Sean Kim

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EDUCATION

Georgia Institute of Technology **Atlanta, GA**
B.S. Computer Engineering *Expected May 2028*

- **Concentrations:** Distributed System & Software Design; Computing Hardware & Emerging Architectures

SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, React Native, Node.js, Express.js, FastAPI, SQLAlchemy, PyTorch, Hugging Face Transformers
Databases: PostgreSQL, MongoDB, MySQL, SQLite
Developer Tools & Technologies: Git, Docker, GitHub Actions, AWS S3, Linux, REST APIs, GraphQL, CI/CD, TCP/IP

EXPERIENCE

Research Intern **Jun 2025 – Aug 2025**
George Mason University Nanoelectronics Lab *Fairfax, VA*

- **First-authored research poster under Prof. Ethan Ahn** on thermal-treatment optimization for 2D materials, presented at GMU Summer Research Symposium (Aspiring Scientists Summer Internship Program, 11% acceptance rate).
- Increased 2D MoSe2 flake size **~4x (from a 50–70 micron baseline to 270 microns)** by adding a controlled 30–170 degrees Celsius thermal-treatment step to mechanical exfoliation.
- **Benchmarked 3 exfoliation tapes** across 2–4 filtration passes using optical microscopy to identify the highest-yield medium for 2D material preparation.

Co-Founder & Co-Mechanical Lead **Jun 2024 – Jan 2026**
FIRST Tech Challenge Team #26394 — "Mecha Minds" *Clifton, VA*

- Co-founded a 6-person FIRST Tech Challenge team and led mechanical design across two competition seasons, **qualifying for the State Championship.**
- Raised **$10,000+** by securing 9 corporate sponsorships and organizing a community fundraiser to cover costs.
- Designed a modular wrap-around intake and carousel system in OnShape CAD that captured 18+ game pieces in under 2 minutes, iterating assemblies, fastening, and simulating motion to validate mechanisms before fabrication.
- Introduced **100+ children** to robotics by leading 5 STEM outreach events across Fairfax County Public Schools.

PROJECTS

MarketPulse **Sept 2026 - Oct 2026**
Python, FastAPI, PostgreSQL, Docker, PyTorch

- **Engineered** a containerized financial intelligence platform using **Python, FastAPI, PostgreSQL, and Docker**, developing RESTful APIs to retrieve and analyze **10,000+ financial news articles** across **25+ publicly traded companies**, achieving **<150 ms average API response latency**.
- **Developed** an automated **ETL pipeline** integrating the Finnhub API, Pydantic validation, and SQLAlchemy to ingest **500+ articles daily**, implementing PostgreSQL deduplication, error handling, and scheduled ingestion every **15 minutes** to maintain **99%+ data processing success rates**.
- **Integrated FinBERT and VADER sentiment models** using **PyTorch and Hugging Face Transformers** to classify financial headlines, processing **1,000+ articles per hour** and achieving **85%+ classification accuracy** on a labeled financial sentiment evaluation dataset.
- **Designed** a relational PostgreSQL schema for financial articles and sentiment predictions, implementing indexed queries, automated **pytest** testing, and **GitHub Actions CI/CD** to achieve **85%+ test coverage** and reduce average query latency by **40%** through indexing and query optimization.

Oratio **Aug 2024 – Jun 2025**
Python, TensorFlow, Google Colab

- Built a 1D CNN classifying five childhood speech-error patterns at **94.69% accuracy / 94.66%** F1 using TensorFlow on Google Colab.
- Engineered a **novel text-based pipeline** classifying IPA transcriptions instead of audio — bypassing noise, pitch, and privacy limits in prior speech-ML literature — by simulating phonological-error rules (e.g., /k/ to /t/ fronting), transcribing to IPA, then integer-mapping and zero-padding a 563-word dataset.
- Designed the network architecture — **embedding -> Conv1D (64 filters) -> max-pooling -> dense (128-unit ReLU, softmax)** — trained with Adam optimizer and categorical cross-entropy loss.
- Capped training at **20 epochs** against an ~12-epoch accuracy plateau to prevent overfitting; validated with precision, recall, F1, and a confusion matrix to surface per-class errors under class imbalance.